

# VAANEE TRIPATHI

99-7116-2648 | [vaanee.tripathi\\_ug25@ashoka.edu.in](mailto:vaanee.tripathi_ug25@ashoka.edu.in) | [Github](#) | [LinkedIn](#)

[Personal Website](#)

## EDUCATION

**Ashoka University** Sonepat, India  
*B.Sc. in Computer Science* March 2026

**Apeejay School** Noida, India  
*Senior School Examination* March 2021

**Apeejay School** Noida, India  
*Secondary School Examination* March 2019

## TECHNICAL SKILLS

### High Exposure

- Agentic Development, AI Foundry, L<sup>A</sup>T<sub>E</sub>X, MCP Server Programming, Qualitative Research, Rapid Prototyping, Working with Microcontrollers, Python, C, Technical Writing, Prompt Engineering, Git, Computer Vision, Soldering, 3D printing,

### Limited Exposure

- Thematic Analysis, Figma, Canva, Wix, CAD modelling, Data Analysis and Visualisation, PCB Designing, Machine Learning, NLP, HTML, CSS, Jekyll

## SOFTWARE PROJECTS

**Join the Dots** | *A tool to teach visually-heavy concepts to VI students* August 2023 – May 2026

- Developed agentic tool using VLMs, ML models, and OpenCV for real-time accessible multimodal video content presentation to visually impaired users; evolved from frugal refreshable braille display prototype (HCII 2025 short paper & ATIA 2025 poster).
- Conceptualized end-to-end pipeline and individually coded OpenCV-based video data extraction for hardware integration, optimizing for classroom real-time use.

**NVDAI** | *Making non screen-reader friendly websites accessible* July 2025 – December 2025

- Led end-to-end development of NVDAI, a screen reader extension built from scratch using MCP server-client architecture to enable AI-powered computer use and visual UI understanding for visually impaired users
- Architected end-to-end system design, self-learning Model Context Protocol to implement robust client-server communication for real-time accessibility and intelligent UI interpretation

**Accessible Intro to CS Curriculum** | *Published: ACM COMPUTE'25 (full paper)* February 2025 – July 2025

- Devised a comprehensive 5-pronged strategy to improve accessibility in CS education, addressing content, visual aids, inclusive assignments, psychosocial support, and teaching team training.
- Conducted interviews with students with VI, designers, professors, professionals, and learning support staff to incorporate real-world insights into the curriculum design.
- Designed a tactile learning kit to translate visual computing concepts into accessible, multisensory formats.
- Preparing for user testing with visually impaired CS students to collect feedback and iterate on the curriculum for real-world effectiveness for a course with 100+ students

## HARDWARE PROJECTS

**The Air Purifier Project** | *A frugal open-source device to make clean air accessible* February 2025 – August 2025

- Conducted extensive research on air purification principles and fluid dynamics to inform structural design and airflow optimization.
- Created a comprehensive moodboard and design brief to guide frugal, open-source development.
- Currently designing and building a functional prototype of the air purifier, focusing on low-cost materials and efficient construction as well as a sensor-based system to validate filtration efficacy.

**Acoustic Levitation and Holography** | *using sound for levitation and holography* June 2023 – November 2023

- Proposed the project to the department, received funding and took it up individually as a freshman
- Gained an understanding of the complex physics, mathematics, electronics involved - picked up custom PCB designing, an understanding of FPGAs, and read research papers for the same
- Curated a list of requirements for the project, found vendors across three countries and co-ordinated with them

**Generative Art Project** | *a novel medium of performance using projections* January 2023 – April 2023

- Developed and executed a unique project idea at the intersection of art, performance, and technology.
- Learnt a new software within a two-month time-frame and gained enough proficiency to help other team members through the process.
- Conducted thorough hardware research, aided in handling logistics, and conceptualized creative performance themes and script.

## WORK EXPERIENCE

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### Research Intern

June 2026 - October 2026

*Project Ekalaiva*

*Microsoft Research India*

- Working under **Dr. Manohar Swaminathan** to realise a six-pillar framework to transform education
- Leading a multi-institutional user study to study impact, derive insights, and study classroom behaviour
- Developing an persistent AI multi-agentic personal assistant
- Designing novel interfaces for these assistants from scratch.

### Research Intern

May 2025 - August 2025

*I-STEM + University of Michigan*

- Contributed as lead developer on NVDAI, an AI-powered accessibility tool (see Projects)
- Collaborated with IDEA Lab at University of Michigan in collaboration with I-Stem on assistive technology prototyping and research methodology (Under Professor Venkatesh Potluri)

### Teaching Assistant

August 2025 - May 2026

*Ashoka University*

- Assisted in teaching a class of 60+ students Design Practices in Computer Science
- Assisted in teaching an Introduction to Computer Science class of 100 students
- Conducted routine office hours, corrected assignments, and helped create a technically rigorous and safe space for learning

### Makerspace Intern/Community Mentor

June 2023 - May 2025

*Mphasis Ashoka Digital Makerspace*

*Ashoka University*

- Led design of comprehensive makerspace curriculum for undergraduate course and organized accessible knowledge repository; contributed to apparatus design, software development, and research consultation for 5+ professors and 6+ projects, including NCBS prototyping.
- Mentored high-school/PhD students, trained 30 undergraduates in hardware/software workshops, and enhanced operations

### Mentor

September 2024 - Present

*CreateED*

*Ashoka University*

- Guided students in electronics, 3D printing, and software design, supporting them in developing innovative projects for prestigious competitions, including IRIS.

### Project Intern

June 2024 - July 2024

*Indian Institute of Technology*

*Bombay*

- Participated in a six-week intensive invention program developing a product to monitor food health and provide early spoilage alerts. Provisional Patent pending
- Collaborated with peers to brainstorm, research, design, and prototype innovative solutions addressing real-world challenges.

## COMMUNITY & LEADERSHIP

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### President

August 2023- Present

*Women in Computing Society*

*Ashoka University*

- Organised the nation's first student-led accessibility and inclusivity tech conference with 200+ registrations, nation-wide and international speakers
- Led and managed a team of 40 members, creating a collaborative and inclusive space and facilitated events like cryptic hunts, workshops, alumni mentorship programs and more for enhancing the CS culture

### Lead on Design and Logistics

June 2023 - August 2023

*RedBrick Hacks*

*Ashoka University*

- Learned new design software in two months and created majority of the collaterals and led a team of 10.

- Handled the logistics for an event with 200 participants and 400 applications involving but not limited to communication between sub-teams as well as diff university depts and geographically distributed team members

## Point of Contact

November 2021 – Present

*Research Reading Group*

*WiCS × CS Society, Ashoka University*

- Established the group, managed logistics, student discussions, and paper selection - which facilitated weekly research paper discussions presented by fellow students.

## RELEVANT COURSEWORK

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Frugal Science (Stanford University)  
 Design Practices in CS  
 Trustworthy AI  
 Rapid Prototyping and Experimentation  
 Design and Development of an Accessible CS curriculum  
 Computer Vision  
 Natural Languages Processing  
 Principles of Science  
 Introduction to Critical Thinking  
 Storytelling in the Digital Age

## PUBLICATIONS

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**Accessibility Beyond Accommodations: A Systematic Redesign of Introduction to Computer Science for Students with Visual Impairments** | *ACM COMPUTE*, 2025

This paper presents a comprehensive framework for re-designing university-level introductory CS curricula to provide equitable learning experiences for students with visual impairments without requiring specialized technical infrastructure.

**Dots: A Frugal Refreshable Tactile Dot Grid for Students with Visual Impairments** | *HCI*, 2025

The paper presents an approach to employ low-cost hardware components, open-source computer vision algorithms, and a modular design to create an affordable alternative to commercial solutions to traditional RTDGs